

Shayan Shakeri

shayansnezhad@gmail.com | (416) 471-6651 | GitHub | merlyn-labs.com

Research

AI Researcher & Co-Founder, Merlyn Labs – Toronto

Aug 2025 – Present

- Training VLAs via RL in OmniGibson simulator; using VLM judges to generate dense reward functions per task
- Built runtime monitoring into live VLA rollouts to detect behavioral anomalies and trigger automated re-planning
- Extended open-source RLinf distributed RL framework to support flow-matching VLA training in OmniGibson
- Studying Sim2Real transfer from OmniGibson using in-house AlohaMini robot

Stanford BEHAVIOR-1K Challenge – 8th Place, Standard Track

Sep 2025 – Nov 2025

- Identified proprioceptive collapse as critical failure mode in flow-matching VLAs, improving task success by up to 48%
- Mitigated policy collapse via 60% proprioceptive masking; published model organisms analysis on LessWrong
- Found 3x gain from chunking over temporal ensembling, exposing temporal awareness vulnerability in VLA architectures
- Published full technical report at : Removing Bottlenecks in VLA Training

Experience

Software Developer – GenAI, Bank of Montreal (BMO) – Toronto

Sep 2025 – Apr 2026

- Developed a GenAI tool hosted on AWS Bedrock converting investment reports into wealth management dashboards
- Designed LLM alignment evaluation for financial outputs; uncovered systematic bias to downplay investment risk
- Tested across multiple models with hundreds of evaluation cases; found bias became visible only at scale
- Building agentic LLM capabilities into Snowflake data platform for automated data search and organization

Biomedical & Electrical Engineering Co-op, Epineuron Technologies – Mississauga

May 2021 – Aug 2022

- Helped develop PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials
- Designed and assembled **PCBs** and authored validation protocols (IEC 60601-1, ISO 13485) for clinical test runs
- Optimized power draw via oscilloscope/power analyzer testing, achieving **900%** battery life improvement
- Modeled electromagnetic field penetration into peripheral nerves in **COMSOL**; data used to optimize electrode diameter

Projects

BardSong, AI-Powered D&D Storytelling Tool

Jun 2025 – Present

- Built end-to-end pipeline: Groq speech-to-text → Gemini scene extraction → image generation → narrated video
- Fine-tuned LLM on D&D sourcebooks for improved narrative coherence; currently in closed alpha with **23 DMs**

Voice-Controlled Robotic Prosthetic, University of Toronto

Sep 2024 – Mar 2025

- Developed LLM-based control pipeline converting natural language to manipulation sequences for 7-DOF simulated arm
- Integrated YOLO object detection with LiDAR depth mapping for 3D localization and grasp planning

VP Technical, McMaster Medical Design Team

Apr 2018 – Apr 2023

- Led development of open-source motorized prosthetic for lower-arm amputees that anyone could assemble for **<\$300**

Education

M.Eng, MIE – AI & Robotics, University of Toronto (Anticipated Graduation: 04/27)

Sep 2024 – Apr 2027

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics Co-op, McMaster University – Dean's Honour List

Sep 2018 – Jun 2023

Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems

Skills & Interests

SFWR & ML: Python, C/C++, PyTorch, Reinforcement Learning, Deep Learning, Fine-tuning, Agent Evaluation

Robotics: VLA Models (π_0 , $\pi_{0.5}$, OpenVLA), OmniGibson, Isaac Sim, MuJoCo, Sim-to-Real

Hardware: PCB Design, COMSOL, SolidWorks, Oscilloscopes, 3D Printing, Soldering, Machining

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano